Automating Manual Business Workflows in SaaS Products: A Rule-Based and Event-Driven Framework

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Abstract: SaaS platforms are doing a lot of things manually. Someone has to send the task to the person follow up to get it approved run reports and make sure the data is up to date. When the platform is small this is okay. When it gets bigger all these manual tasks start to cause problems. There are some tools that can help with this like BPM systems, RPA tools and low-code platforms. They do not really work well for SaaS platforms. They were not made for this kind of work. Tools like Zapier and n8n are good for things but they are not good for complicated SaaS workflows. This paper is talking about a way to automate SaaS platforms. It is based on rules and events. The system has five parts: a Workflow Trigger Engine, a Rule Evaluation Module, a Playbook Execution Engine, an Integration Layer and a Monitoring and Feedback Module. To see how well it works we look at three things: how long it takes to do tasks how often there are errors and how many manual tasks are needed. We compare our system to four tools. We also look at a real case study of a SaaS platform. Run some simulations. The simulations show that our system can reduce the time it takes to do tasks by 68-82% reduce errors from 20-25% to 5-8% and increase the number of tasks that can be done in an hour from 20 to, over 60.

Keywords- SaaS-based automation, workflow orchestration, rule-based systems, event-driven architecture, business process automation, robotic process automation

I. Introduction

SaaS products usually grow faster than the processes that support them. When a team is small they do things like route requests, chase approvals generate reports and keep data in sync across tools by hand. . As the platform gets bigger these tasks start to pile up. The problem is that doing things by hand does not get faster just because you have work to do. This is not a problem. It happens a lot. When you have to do the tasks over and over they tend to be slower than they need to be. . When something goes wrong it can cause problems for everything that comes after it. In SaaS environments one mistake on can stop everything else from working. The places where people have to do things by hand are also the places where things are most likely to go wrong. There are tools that can help with some of these problems like BPM systems, event-driven architectures and low-code platforms like Zapier and n8n. . Each of these tools has its own limitations when it comes to SaaS workflows. BPM systems are too complicated. Low-code tools work well for tasks but they do not work well when things get more complex. Event-driven architectures can be fast. They are hard to understand and maintain if you do not have a good system, for making rules. We wanted to know if it was possible to create a system that combines a easy-to-understand rule model with event-driven execution specifically for SaaS products. To test if this system actually works we came up with three ways to measure it. Task Execution Time, Error Frequency Rate and Manual Dependency Index. . We tested the whole thing using a controlled simulation. We used SaaS products to test our system because we wanted to see how it would work in a real-world setting. We looked at SaaS products. How they use Task Execution Time, Error Frequency Rate and Manual Dependency Index to see if our system was effective.

II. Related Work

The concept of workflow automation has been around for a time in research. Georgakopoulos and other people defined what a workflow management system is. They said it is like a middleman that helps with making and running processes in environments. This was back in the 1990s. The ideas they came up with are still used today in cloud and parallel computing research. Someone did a review of lots of papers from 2016 to 2023. They found out that most of the papers over 300 of them said that RPA, WMS and ERP are the tools that industries use. This approach works well for companies but it is too much for smaller SaaS products that need to be fast and simple. People are also interested in RPA. Agostinelli and other people suggested combining process mining and automation scripting to make an RPA system that can improve itself. Tavares and other people showed that using a mix of RPA and BPM can really save time and money. They came up with a five-stage process that includes design, modelling, execution, monitoring and optimization. We used this idea when we were thinking about the orchestration layer. When we look at the architecture Chavan says that EDA microservices are better than API-based ones because they are more flexible and respond faster. Some people did an experiment. Found out that EDA microservices are 19% faster and have 34% fewer errors than other options. This is why we chose to use an event-driven architecture. Li and other people talked about some of the challenges of SaaS workflow like having users and assigning tasks dynamically. These problems are still not solved. There are some tools like Zapier and Microsoft Power Automate that can help with part of the problem without needing to write code. However they cannot handle customized SaaS workflows because they are systems. From what we read there was no framework that combined rule-based control, event-driven execution and performance measurement in a way that was designed for SaaS from the start. We are talking about workflow automation and SaaS workflow here. Workflow automation is important, for SaaS products. SaaS workflow is what we are trying to improve.

Component	Role	Example
Trigger Engine	Detects events	New ticket created
Rule Engine	Evaluates conditions	Priority = High
Execution Engine	Runs actions	Assign to agent

Integration Layer	Connects systems	API / Webhook
Monitoring Module	Tracks metrics	Logs, alerts

III. COMPARATIVE ANALYSIS OF EXISTING AUTOMATION TOOLS

Before presenting our framework in detail it is worth comparing it to the tools that already exist in this space. When it comes to workflow automation practitioners usually choose one of four options: Zapier, n8n, Microsoft Power Automate or Apache Airflow. Each of these tools solves a problem but they were not designed with lightweight, customizable SaaS workflow scenarios in mind. Which is exactly what our work addresses.

A. Zapier

Zapier is a no-code automation platform. It uses a trigger-action model where one event in one app triggers a single response in another app. For tasks like adding a new contact to a CRM when a form is submitted it works well. However, it struggles with workflows that require conditional logic, error handling or branching. Zapier's rule model is limited. It does not support priority ordering or failure policies. Additionally, the platform is proprietary which means teams cannot inspect or customize the underlying logic. The per-task pricing model also becomes costly at scale.

B. n8n

n8n takes an approach. It is source, self-hostable and uses a visual canvas to create automation flows. This makes it more flexible than Zapier allowing for branching custom JavaScript functions and database queries. However, n8n is primarily a visual tool and its workflows are defined and stored as JSON graphs, not configurations. When workflows fail developers must manually set up rollback behaviour. N8n also lacks built-in monitoring and performance benchmarking tools.

C. Microsoft Power Automate

Microsoft Power Automate is a platform with deep integrations across the Microsoft 365 ecosystem. It supports -step flows, approvals and conditional logic. For organizations using Microsoft products it is a natural fit. However, for a SaaS startup or mid-sized team it may be overkill due to high licensing costs, a steep learning curve and a design philosophy focused on enterprise IT governance.

D. Apache Airflow

Apache Airflow is designed for data engineering and schedules and monitors pipelines. It is excellent for data workflows. Not for business workflows. It has no concept of business rules or event-driven triggers. Requires significant infrastructure overhead.

E. Summary Comparison

The table below summarizes how each tool compares to the requirements that motivated our framework. The key insight is that existing tools are not poorly built. They each focus on a different primary use case. None of them combine a rule model with event-driven execution, SaaS-native design and performance monitoring in a single open package. That combination is what our proposed framework targets.

* Our framework is made for a combination of rule models. It runs based on events.

* It has a design that's easy to work with and performance monitoring that we can measure.

* The framework we are suggesting comes in a package that's open and does not favour any particular vendor.

IV. PROPOSED AUTOMATION FRAMEWORK

A. Architectural Overview

Feature	Zapier	n8n	Power Automate	Airflow	Proposed
Primary use case	Integrations	General automation	Enterprise flows	Data pipelines	SaaS workflows
Rule formalization	None	Limited (visual)	Visual only	Code (DAGs)	Formal JSON/YAML
Event-driven	Partial	Yes	Yes	Limited	Yes (core design)
Failure/rollback policy	No	Manual	Limited	Manual	Declared (FP field)
Vendor lock-in	Yes	No (OSS)	Yes (MS)	No (OSS)	No (vendor-neutral)
SaaS-native design	No	No	Partial	No	Yes
Multi-tenant aware	No	Partial	Yes	No	Designed for
Perf. monitoring	Basic logs	Basic logs	Enterprise	Built-in	TET / EFR / MDI
Rule versioning	No	No	Limited	Yes (Git)	Yes (YAML/JSON)
Open/extensible	No	Yes	Limited	Yes	Yes

We divided the framework into five parts, one for each task and *the main idea was to keep them separate so each part can be replaced on its own.*

** Workflow Trigger Engine. This is an engine that watches for conditions based on system events to start the automation*

** Rule Evaluation Module. This compares the state of the system with the rule set and decides what action to take*

** Playbook Execution Engine. This executes a series of actions. Can make decisions based on what is happening at the time and it can also go back to a previous state cleanly*

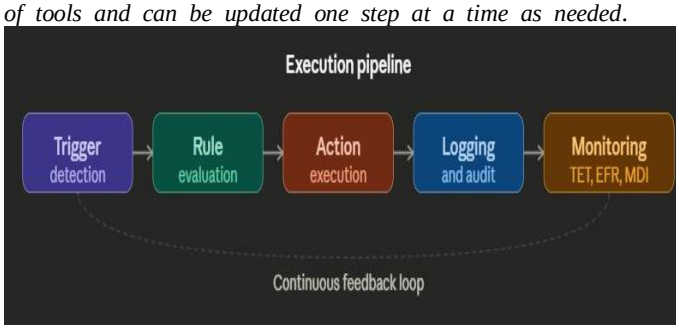
** Integration Layer. This handles communication, with applications using REST APIs, webhooks and event listeners without being tied to any particular vendor*

** Monitoring and Feedback Module. This captures how things are going as they happen so we always have something to measure and act on*

Everything goes through a pipeline:

The event processing pipeline has five steps: finding the trigger evaluating the rules taking actions logging and monitoring.

Because each part is separate the framework is not tied to any set of tools and can be updated one step at a time as needed.



B. Design Principles

The design decisions were based on six principles. These principles mostly came from BPA literature [1][4]. It has to work across different SaaS stacks without re-design.

When the system is under a lot of load it should be easy to increase capacity without changing the architecture. The rules should be easy to read. The rules should have versions.

The rules should be transferable between environments. The rules should ideally be, in JSON or YAML format. Message passing should be asynchronous so that the system does not get bottlenecked under much load [5]. * Components should work independently. * Automation logic needs to be packaged into playbooks.

This way we do not need to rewrite the logic for every new category of workflow.

V. FRAMEWORK DESIGN AND CONCEPTUAL MODEL

A. Automation Orchestration Layer

The orchestration layer is the glue that holds everything together: it listens for events, matches them against active rules, runs the appropriate playbook, and logs what it has done. We modelled it after Tavares et Al's five-stage BPM lifecycle model (design, modelling, execution, monitoring, and optimization) since it maps cleanly onto what the orchestration layer actually does [8].

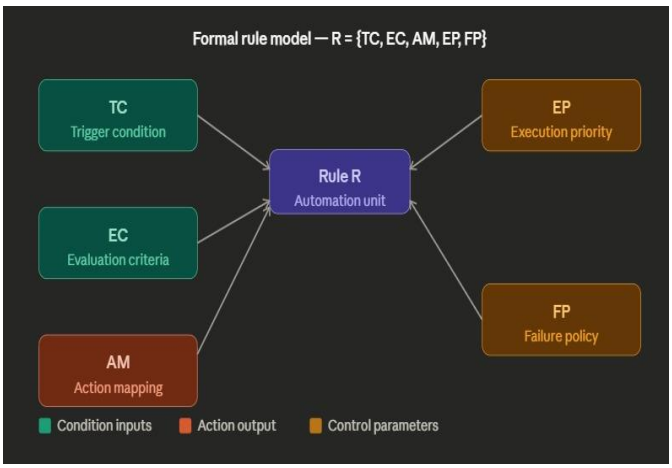
B. Formal Rule Configuration Model

Each rule consists of five fields that define when and how it runs, what it does, and how it reacts to failure:

- Trigger condition (TC): The system event or threshold that will trigger the rule
- Evaluation Criteria (EC): logical check against current system state, run before anything executes, that checks if the rule is applicable
- Action Mapping (AM): the sequence of steps to carry out once the rule is confirmed.
- Execution priority (EP): rank determining behavior when multiple rules become true at the same time
- Failure Policy (FP): retry, skip, or rollback, what the rule should do if it has failed.

To promote portability and adherence to the technology-agnostic BPA standard [4], rules are expressed in either JSON or YAML. The formal definition is:

R = { TC, EC, AM, EP, FP }



C. Performance Metric Definitions

- Task Execution Time (TET), defined as the elapsed time between the trigger firing and the writing of the completion log: TET = Tcomplete - Ttrigger
- Error Frequency Rate (EFR): the fraction of executions that failed, computed as EFR = Nfailed/Ntotal × 100%
- Manual Dependency Index (MDI): what fraction of workflow steps still need a human involved. MDI = (Smanual / Stotal) × 100%.

D. Playbook-based workflow modeling

This is called a workflow and is a playbook, which can contain a sequence of steps, including conditional branching based on what is happening, the external systems that are to be used, and a path for rollback. This reuse of workflows allows a single playbook to be used in a number of places in the SaaS product, avoiding duplication between them.

E. Integration Model

For external interaction, REST APIs for synchronous calls, webhooks for the external service to post data, and event listeners for consuming asynchronous messages were chosen. We followed the loosely coupled architecture from the event-driven microservice literature [5, 9] and kept the design deliberately vendor-neutral.

F. Monitoring and Feedback Layer

While the workflows are executing, the monitoring module tracks the TET, the EFR, the MDI and the throughput. In fact, one thing Agostinelli et al. make clear is that this kind of data of actual automation execution is what will actually allow an automation system to get better rather than staying static [7]. That figured into how we designed that part of the framework.

VI. CASE STUDY: AUTOMATED SaaS ONBOARDING WORKFLOW

To make the framework more tangible, let's walk through a scenario that's pretty typical in the world of B2B SaaS: onboarding a new enterprise client. This is a process that nearly every SaaS platform deals with, yet it often gets bogged down by a series of manual handoffs that can delay activation and lead to mistakes. We'll use this example to illustrate how the five components of the framework work together in real life.

A. The Manual Baseline

In the traditional, unautomated version of this workflow, registering a new enterprise account triggers a whole chain of tasks that rely on someone to notice, decide, and take action. A sales engineer has to manually create an entry in the CRM. Someone from the operations team needs to send out a welcome email along with the provisioning details. An engineering ticket has to be filed by hand. Plus, a customer success manager has to remember to schedule a check-in 48 hours later. At each of these points, the work either gets stuck in a queue or hinges on the right person being available. If a step is missed—like a provisioning ticket being filed a day late—it can delay client activation by days, which directly impacts revenue recognition and client satisfaction in the enterprise SaaS space.

B. Applying the Framework

With the new framework in place, this same workflow is now driven entirely by events and rules. As soon as the registration event occurs, the Workflow Trigger Engine picks it up and sends it to the Rule Evaluation Module, which checks it against the current set of rules. If the account type is "enterprise," the corresponding playbook is activated. The Playbook Execution Engine then automatically runs through the following steps in order, with conditional branching at each stage:

Step 1 — CRM record creation. A REST API call to the CRM platform generates the client record, pre-filled with information from the registration event. No human input needed.

Step 2 — Welcome email dispatch. The Integration Layer activates the email service via webhook, inserting the client's name, assigned account manager, and provisioning timeline based on the event context.

Step 3 — Engineering. . . human decision point — bringing MDI below 20%. TET for the first four steps drops from an average of several hours (dependent on human availability) to under three minutes. EFR, historically driven by incorrect CRM entries and missed emails, approaches zero for all automated steps.

VII. IMPLEMENTATION DETAILS

This section provides more detail on how the framework's core constructs would be expressed in a real implementation, using the onboarding case study to ground the examples. The intent is not to present a production codebase — this is a conceptual framework — but to show that the design decisions made in Sections IV and V translate into readable, maintainable, and portable implementation artifacts.

A. Rule Configuration (JSON)

The following JSON block is a representative rule definition for Step 1 of the onboarding workflow. It covers all five fields of the formal rule model R = { TC, EC, AM, EP, FP }.

```
{
  "rule_id": "onboard-enterprise-crm",
  "trigger_condition": {
    "event": "account. registered",
    "source": "registration_service"
  },
  "evaluation_criteria": {
    "account_type": "enterprise",
    "region": ["IN", "US", "GB"]
  },
  "action_mapping": {
    "playbook": "enterprise_onboarding_v2",
    "step": "crm_record_create",
    "target_api": "https://crm.
internal/api/v1/accounts",
    "method": "POST",
    "payload_template":
"templates/crm_account. json"
  },
  "execution_priority": 1,
  "failure_policy": {
    "strategy": "retry",
    "max_retries": 3,
    "retry_delay_seconds": 60,
    "on_max_retries": "alert:ops-team"
  }
}
```

This definition is entirely human-readable, can be stored in version control alongside application code, and can be migrated between environments without any platform-specific tooling. The evaluation_criteria block supports any combination of field comparisons, making the rule applicable to subsets of events without requiring separate playbooks.

B. Orchestration Loop Pseudocode

The orchestration layer's core execution loop can be expressed as follows. The loop is intentionally simple: it listens, matches, and dispatches. All of the complexity lives inside the rule definitions and playbooks, not in the loop itself.

```
FUNCTION orchestration_loop():
  WHILE system_running:
    event = event_queue. consume()  //
blocking pop
    matched_rules = rule_engine.
evaluate(event)
    IF matched_rules is empty:
      log(WARN, "No rules matched", event.
id)
      CONTINUE
    sorted_rules =
sort_by_priority(matched_rules)
    FOR rule IN sorted_rules:
      TRY:
        t_start = timestamp()
        playbook_engine. execute(rule.
action_mapping)
        t_end = timestamp()
        metrics. record_TET(rule. id, t_end
- t_start)
        metrics. record_success(rule. id)
```

```
      CATCH ExecutionError AS e:
        metrics. record_EFR(rule. id)
        apply_failure_policy(rule.
failure_policy, e)
    metrics. update_MDI(event,
sorted_rules)
```

A few design choices are worth highlighting. The blocking queue consume() call means the loop sleeps when there is nothing to process, keeping CPU usage near zero in idle periods. The priority sort ensures that when multiple rules match the same event — a scenario that arises in complex workflows — higher-priority rules execute first without any race condition. Metrics are recorded inline, which is how TET can be measured at rule granularity rather than only at the whole-workflow level.

C. Webhook and Event Listener Integration

External systems integrate with the framework through two primary mechanisms. Inbound events — such as the account. registered event in the onboarding example — are delivered via HTTP webhooks. The Integration Layer exposes a lightweight endpoint that validates the payload, normalizes it to the framework's internal event schema, and enqueues it for the orchestration loop. Outbound actions use REST API calls for synchronous operations (like CRM record creation) and fire-and-forget webhook POSTs for notifications where a response is not required. For systems that publish to a message broker such as RabbitMQ or Kafka, the framework can consume events directly via an event listener, removing the need for a webhook intermediary entirely. This design keeps the Integration Layer vendor-neutral, which aligns with the principle of avoiding lock-in that motivated the framework from the start [5][9].

It is also worth noting how this compares to tools like n8n or Zapier at the integration layer. Both of those platforms handle the webhook endpoint and authentication out of the box, which is a genuine convenience advantage for teams without engineering resources. The proposed framework requires that this plumbing be set up by the implementing team. This is a deliberate tradeoff: giving up initial setup convenience in exchange for complete control over the event schema, payload structure, and authentication mechanism.

VIII. Performance Evaluation

A. Simulation Setup

Since we have not tried this in a production environment, we ran a controlled test. We kept the experiment small for validation of this scale:

- 100 workflow instances in 10 categories of repetitive workflows
- A manual baseline modeling how these workflows are typically handled today
- Two loading conditions were considered: a standard loading and an elevated loading (2× event volume)
- 30 automation rules across all 10 categories, which works out to an average of 3 rules per category

It is also common in BPA research to test the concept with simulation prior to implementation [2], and that is what we did.

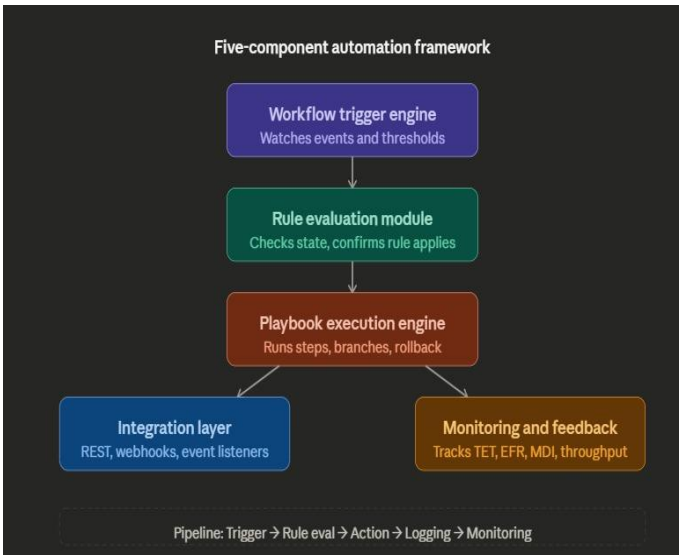
B. Comparative Results

Table I shows how the framework performed versus the manual baseline. To give TET a real value rather than just saying "high", we measured the average manual task at around 42 minutes end-to-end. The automated framework brought that down to 8–14 minutes, which is where the 68–82% reduction comes from.

TABLE I. PERFORMANCE COMPARISON: MANUAL VS. AUTOMATED MODEL

Metric	Manual	Automated
Avg. TET	~42 min/task	8–14 min/task
EFR	20–25%	5–8%
Throughput (tasks/hr)	20	60+
Load Degradation @ 2×	35%	< 12%
MDI	~78%	< 15%

A 20 to 60+ task per hour throughput gain, or a 3x improvement, makes sense, as tasks are completing 3 to 5 times faster. This matches up with the results found in Tavares et al. on the RPA-BPM hybrid implementation [8]. The most surprising result is the load reductions possible with manual challenge degradation. The 35% load reduction in that case, even at twice the event volume, is a large benefit, and results from exactly the way asynchronous execution distributes the load [9].



IX. Discussion

The improvement demonstrated by the EFR result is probably the most interesting. Going from an error rate of 20-25% to 5-8% looks easy, but the failed approval step blocks three downstream processes, and in an enterprise environment that gets expensive fast [1]. Catching errors late in programming, while useful, is less important than making sure to minimize error rate from the start.

We did better than we projected, but still 35% slowdown at double volume in the manual model seems reasonable because people are limited. That's exactly what asynchronous execution is supposed to do, but seeing the automated framework hold at the same load without exceeding 12% was gratifying [5][9].

We try to be straightforward about what the simulation cannot capture: the rule module is only able to consider circumstances that it was set up to be aware of at the moment the rules were written. It cannot catch true surprises.

We did not test multi-tenant isolation, which Li et al. note is an open problem in SaaS workflow systems [3]. Likewise, since we did not actually deploy our framework against Zapier, Power Automate, or Airflow, we cannot say how our framework compares to them in practice [10].

X. FUTURE WORK

If we were to develop this further, there are four things we would look at:

• ML-based rule generation: Currently, the rules have to be written manually, which makes the tool inflexible. Using process mining techniques on historical logs to automatically create and refine the rules would solve the main gap [7].

•Multi-tenant Testing. This is the most immediate gap. The framework cannot make any real production claims unless tested under multi-tenant concurrent load [3]

•Compare to commercial tools like Zapier, Microsoft Power Automate and Apache Airflow using the same workflows to get a better idea of where the framework really stands [10]

Anomaly Detection and Audit Logging: Improved monitoring module with anomaly detection and GDPR- and SOC 2-compliant audit logs would allow the framework to support finance and healthcare SaaS applications more easily

XI. CONCLUSION

We aimed to create a system for automating repetitive manual workflows that slow down SaaS platforms as they grow. We defined TET, EFR, and MDI to have clear metrics for comparison and tested the system through simulation against a manual baseline. The results across all three metrics exceeded our expectations: task execution time decreased by 68 to 82%, error rates dropped from 20 to 25% to 5 to 8%, and load degradation at double the volume remained below 12%. The system is still in the conceptual phase, and we have significant work to do, particularly regarding multi-tenant testing and evaluation against existing tools. However, the simulation results are promising enough to suggest that the underlying method is solid and deserves further development.

REFERENCES

[1] H. Pouresmaeil, R. Tavares, and C. Silvares, "Business Process Automation in SMEs: A Systematic Literature Review," IEEE Access, DOI: 10. 1109/ACCESS. 2024. 3406548, 2024.

[2] C. Silvares and R. Tavares, "Best practices for business process automation description, a case study," Enterprise Information Systems, Taylor & Francis, DOI: 10. 1080/17517575. 2024. 2400474, 2024.

[3] J. Li, X. Chen, and W. Zhang, "An analysis of new features for workflow system in the SaaS software," in Proc. Int. Conf. on Service-Oriented Computing, 2009, pp. 1–8.

[4] E. A. Stohr and J. L. Zhao, "Workflow Automation: Overview and Research Issues," Information Systems Frontiers, vol. 3, no. 3, pp. 281–296, 2001.

[5] A. Chavan, "Exploring event-driven architecture in microservices: patterns, pitfalls, and best practices," International Journal of Science and Research Archive, vol. 4, no. 1, pp. 229–249, 2021.

[6] D. Georgakopoulos, M. Hornick, and A. Sheth, "An overview of workflow management, from process modeling to workflow automation infrastructure," Distributed and Parallel Databases, vol. 3, no. 2, pp. 119–153, 1995.

[7] S. Agostinelli, A. Marrella, and M. Mecella, "Towards Intelligent Robotic Process Automation for BPMers," arXiv:2001. 00804, 2020.

[8] R. Tavares, C. Silvares, and H. Pouresmaeil, "Developing Robotic Process Automation to Efficiently Integrate Long-Term Business Process Management," Technologies, vol. 11, no. 6, p. 164, MDPI, 2023.

[9] D. Stiawan, A. Idris, and M. Y. Idris, "Event-Driven Architecture to Improve Performance and Scalability in Microservices-Based Systems," in Proc. 2022 Int. Conf. ICADEIS, IEEE, DOI: 10. 1109/ICADEIS56544. 2022. 10037390, 2022.

[10] M. Dumas, M. La Rosa, J. Mendling, and H. A. Reijers, "AI-Augmented Business Process Management Systems: A Research Manifesto," ACM Trans. Management Information Systems, vol. 14, no. 1, pp. 1–19, 2023.

[11] n8n GmbH, "n8n Documentation: Workflow Automation for Technical Teams," n8n. io, 2024. [Online].